Chapter 5
Relational Database Design
(Normalization)

Normalization

- Decomposition of tables to reduce or eliminate redundancy in database
- Due to redundancy following problem occurs
  - Insertion anomalies
  - Update anomalies
  - Delete anomalies

ENO	ENAME	E_PHONE	DID	DNAME	Location
E-101	Amit Singh	7658943654, 8765487690	D-1	HR	Mumbai
E-102	Ram U.	9654367834, 6789067543	D-2	Research	Delhi
E-103	Riya Joshi	8769875436	D-1	HR	Mumbai

Insertion anomalies-If we want to add new department then we have to add employees data in the table Or if we want to add new employee in already existing department then we have to duplicate information about that department

Delete anomalies-if we want to delete Employee Ram's data then we will loose information about department D-2

Update anomalies-if we want to update the location of D-1 department then it forces to update multiple records

Normal Forms

- 1NF
- 2NF
- 3NF
- BCNF
- 4NF
- 5NF

1 NF

- A relation is said to be in first normal form (1NF) if the value of each attribute contains only a ***single value(atomic)*** from that domain.
- There are **no repeating groups** in the table
- It should not have composite/multi-valued

In layman's terms. it means every column of your table should only contain *single value*

Not in 1NF

ENO	ENAME	E_PHONE	DID	DNAME	Location
E-101	Amit Singh	7658943654, 8765487690	D-1	HR	Mumbai
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1NF

ENO	ENAME	E_PHONE	DID	DNAME	Location
E-101	Amit Singh	7658943654	D-1	HR	Mumbai
E-101	Amit Singh	8765487690	D-1	HR	Mumbai
E-102	Ram U.	9654367838	D-2	Research	Delhi
E-102	Ram U.	6789067543	D-2	Research	Delhi
E-103	Riya Joshi	8769875436	D-1	HR	Mumbai

But problem of primary key which cannot accept duplicate values
Solution --Decompose table by removing multi-valued attribute

ENO	ENAME	DID	DNAME
E-101	Amit Singh	D-1	HR
E-102	Ram U.	D-2	Research
E-103	Riya Joshi	D-1	HR

ENO	E_PHONE
E-101	7658943654
E-101	8765487690
E-102	9654367838
E-102	6789067543
E-103	8769875436

Functional Dependency(FD)

- Redundancy is often caused by a functional dependency
- A functional dependency (FD) is a link between two sets of attributes in a relation

Example of FD

- There exists a functional dependency between A and B ($A \rightarrow B$), if whenever two rows of the relation have the same values for the attributes A, then they also have the same values for all rows of the attribute B.

- A->B
- B->A(not exist)

A	B
a1	b1
a1	b1
a4	b1
a3	b3

List all functional dependencies satisfied by the relation

A	B	C
a1	b1	c1
a1	b1	c2
a2	b1	c1
a2	b1	c3

- A->B(exist)
- A->C(no)
- B->C(No)
- B->A(No)
- C->A(No)
- C->B(exist)
- AB->C(No)
- BC->A(No)
- AC->B(exist)

- List all functional dependencies satisfied by the relation.

X	Y	Z
X1	Y1	Z1
X1	Y2	Z1
X2	Y2	Z1
X2	Y2	Z1

Super Key and candidate key

- Super Key—The set of attributes which can uniquely identify a tuple in a relation is known a Super key
- Candidate key-The minimal of super key is candidate key

Student(ID,roll_no,name,address)

- Super keys

{ID},{roll_no},{ID,roll_no},{ID,name},
{roll_no,name},{ID,roll_no,name}

- Candidate keys

{ID},{roll_no}

- Primary keys

Candidate key chosen by database administrator

Concepts of FD

Student(ID, course_id,cname,fees}

R(A,B,C,D)AB is candidate key

- Key attribute {AB}
- Prime attribute{A},{B}
- Non-prime attribute{C,D}
- Partial FD{B $\longrightarrow$ C,D}
- Full FD {AB->C}
- Transitive FD {AB $\longrightarrow$ C, C $\longrightarrow$ D}
- Trivial FD { C $\longrightarrow$ B }

2NF

A relation is said to be in 2NF if

- It is in 1NF
- It should not have any partial functional dependency

(Every non-prime attribute should be fully functionally depend on key attribute)

2NF example

Find out whether the following relation is in 2NF?

- Student(<u>ID</u>, sname,<u>course_id</u>,cname,fees)

R(<u>A</u>,B,<u>C</u>,D,E) AC is the candidate key

C->D,E } partial FD

A->B } partial FD

D->E }

Relation R is not in 2NF So decompose it

R1=(<u>C</u>,D,E) now C->D,E is full FD as C is the candidate key for relation R1

R2=(<u>A</u>,B,C) Now A->B is full FD as A is the candidate key for relation R2

3NF

- It should be in 2NF
- There should not be any transitive FD

(Every non-prime attribute should be non-transitively dependent on key attribute)

R1=($\underline{C}$,D,E)

R2=($\underline{A}$,B,C)

C->D,E } full FD

A->B } full FD

D->E } Transitive FD

R1 is not in 3 NF because there exist TFD D->E

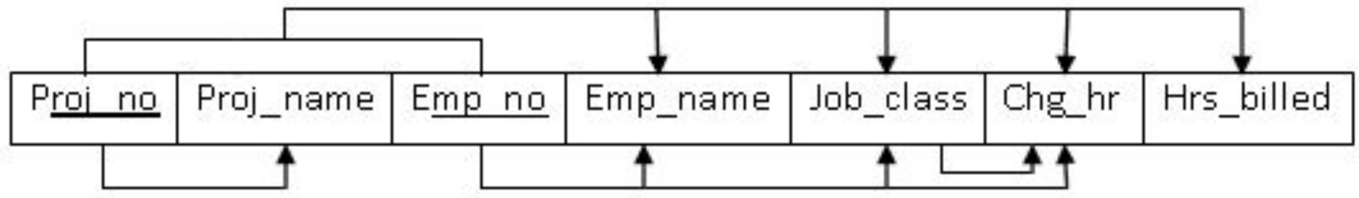
So decompose R1 into R11 and R12

R11=($\underline{D}$,E)

R12=($\underline{C}$,D)

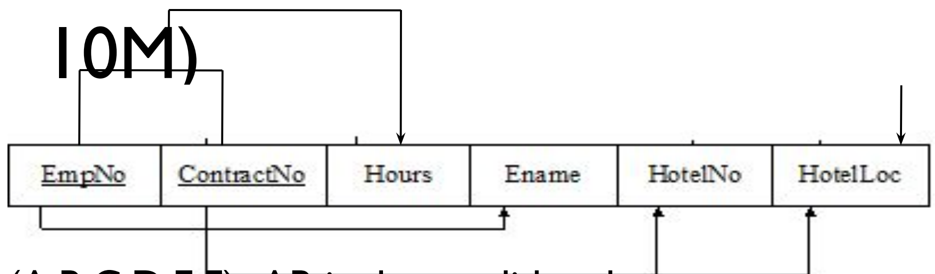
R2=($\underline{A}$,B,C)

Consider a dependency diagram of relation R and normalize it up to third normal form.



R(A,B,C,D,E,F,G)
AC is candidate key
AC->D,E,F,G
A->B
C->D,E,F
E->F
R is in 1 NF as all attributes are single valued
R is not in 2NF because there exist partial dependancies so decompose R in R1 and R2
R1=(A,B) A->B is full FD
R2=(C,D,E,F)
R3=(A,C,G)
R2 is not in 3NF because there is TFD (E->F) so decompose R2 in
R211=(E,F) and R212=(C,D,E)

- Consider the following dependency diagram of relation R and Normalize till 3NF. (May-17, 10M)



R(A,B,C,D,E,F) AB is the candidate key
AB->C
A->D
B->E,F
E->F
R is in 1NF as all attributes are single valued
R is not in 2NF because there exist partial FDs(A->D and B->E,F) so decompose R in R1 and R2
R1=(A,D)
R2=(B,E,F) R3=(C,A,B)
Now all tables are in 2NF. R1and R3 are in 3NF but R2 is not in 3NF so decompose R2 in R21 and R22
R21=(E,F) R22=(B,E)

BCNF

BCNF is the advance version of 3NF. It is stricter than 3NF.

- It must be in 3NF
- for every one of its dependencies $X \rightarrow Y$, at least one of the following conditions hold:
- $X \rightarrow Y$ is a trivial functional dependency ($Y \subseteq X$) OR (AB->B is trivial FD)
- X is a superkey(Candidate kay) for schema R.

Example

TABLE 5.2 SAMPLE DATA FOR A BCNF CONVERSION

STU_ID	STAFF_ID	CLASS_CODE	ENROLL_GRADE
125	25	21334	A
125	20	32456	C
135	20	28458	B
144	25	27563	C
144	20	32456	B

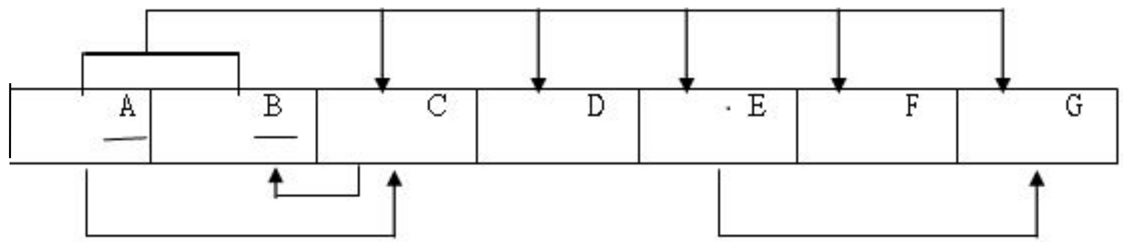
R(stud_id,staff_id,class_code,enroll_grade)

Stud_id,staff_id->class_code,enroll_grade

Class_code->staff_id

Candidate key is Stud_id+staff_id
Above relation is not in BCNF.
WHY?
R1=(class_code,staff_id)
R2=(stud_id,class_code,enroll_grade)

Consider a dependency diagram of relation R and normalize it up to third normal form. (Dec-13, 10M) (Jun-15, 10M). or Consider a dependency diagram of relation R and normalize it up to BCNF normal form. (Dec-17, 10M) (BTL-3)



AB->C,D,E,F,G

A->C

E->G

C->B

Solution:

AB is candidate key

1NF

2NF

R1=(A,C) R2=(A,B,D,E,F,G)

3NF

R21=(E,G)

R22=(A,B,D,E,F)

BCNF

R221=(C,B)

R222=(A,C,D,E,F,G)

Lossless decomposition

- Decomposition is lossless if it is possible to reconstruct relation R from decomposed tables using Joins. The information will not lose from the relation when decomposed. The join would result in the same original relation.
- R=(A,B,C,D,E)
- R1=(A,B,C)
- R2=(D,E,A)

Above decomposition is lossless

- Decomposition of a relation R into R1 and R2 is a lossless-join decomposition if at least one of the following condition satisfies

1. Att(R1) ∩ Att(R2) -> Att(R1) (primary key of R1)

   OR

   Att(R1) ∩ Att(R2) -> Att(R2)(primary key of R2)
2. Att(R1) U Att(R2) = Att(R)
3. Att(R1) ∩ Att(R2) ≠ Φ

Example 1:

R1(<u>A</u>,B,C)

R2(<u>D</u>,E,A)

Above decomposition is lossless decomposition

1. Att(R1) ∩ Att(R2) -> A(Primary key of A)
2. Att(R1) U Att(R2) = Att(R)
3. Att(R1) ∩ Att(R2) ≠ Φ

Example 2:

R1(A,B,C)

R2(D,E)

Above decomposition is lossy as

R1 ∩ R2 ≠ R1

OR

R1 ∩ R2 ≠ R2